

Huffman Algm.

If $T: (w_1 + w_2) w_3 \dots$, then $T_1: w_1, w_2, w_3$

opf> Let $T_2: \text{optimal } w_1, w_2, w_3$
 ~~w_1~~ ~~w_2~~ add $(w_1 + w_2)$

T_3

$$W(T_2) = W(T_3) + w_1 + w_2$$

$$W(T_1) = W(T) + w_1 + w_2$$

$\therefore T: \text{optimal}$

$$\therefore W(T) \leq W(T_3)$$

$$\text{if } W(T) < W(T_3)$$

$$W(T_1) < W(T_2) \quad *$$

Kruskal's Algm.

T : spanning tree by Kruskal's Algorithm

$e_1, e_2, e_3, e_4, \dots, e_k$

optimal: $T': e_1, e_2, e_3, \dots, e_{k'} \dots$

T : Kruskal

T_1 : optimal

$$d(T, T') = r$$

①
②

$$r = n \longrightarrow T_1 = T$$

$$r \leq n-1$$

$T: e_1, e_2, \dots, e_{r-1}, e_r, \dots$

$T_1: e_1, e_2, \dots, e_{r-1}, e_r, \dots$

$T_2: e_1, e_2, \dots, e_{r-1}, e_r, e_{r'} \dots$

$$\therefore W(T_2) = W(T_1) + W(e_r) - W(e_{r'})$$

$$W(e_{r'}) \geq W(e_r)$$

$$\therefore W(T_2) \leq W(T_1)$$

$$W(T_2) = W(T_1)$$